Semantic Onboarding: Ontology-Guided Construction of Formal Knowledge Bases

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Keywords: Semantic Onboarding; Knowledge Engineering; Ontology Alignment; Large Language Models; Executable Knowledge Models; Enterprise Decision Support

Abstract: Semantic Onboarding is an ontology-guided methodology for transforming heterogeneous organizational information artifacts into executable knowledge models for semantically grounded decision support. Traditional enterprise information is often fragmented across unstructured documents and structured operational systems, limiting semantic interoperability and formal reasoning. The proposed methodology combines LLM-based formalization, schema induction, instance-level knowledge generation, and ontology-guided alignment to construct machine-interpretable knowledge bases from heterogeneous sources. Representative artifacts are first transformed into candidate logical models, from which a schema (TBox) is induced to guide the generation of instance-level representations (ABoxes). These are subsequently aligned with a domain ontology, enabling the identification of semantically meaningful relations and executable logical inference. The methodology is evaluated using 72 real-world service contracts and structured operational data from a large public financial institution. Results demonstrate the feasibility of constructing ontology-aligned executable knowledge models that support semantic traceability, logical reasoning, and semantically grounded decision-support processes.

1 Introduction

Organizations increasingly depend on heterogeneous information artifacts to support operational and strategic decision-making processes. These artifacts typically include unstructured documents, such as contracts, reports, and institutional policies, alongside structured operational data stored in relational systems. While such sources contain valuable domain knowledge, much of this information remains embedded in textual descriptions, fragmented across disconnected systems, or represented without explicit semantics suitable for machine interpretation. This limits its semantic interoperability and makes the systematic construction of formal knowledge models for reasoning and decision support particularly challenging.

Recent advances in Large Language Models (LLMs) have created new opportunities for transforming natural language artifacts into structured representations. Existing approaches have shown promising results in tasks such as information extraction, knowledge graph construction, and semantic enrichment. However, many of these approaches remain primarily extraction-centric, producing structured representations intended for retrieval, indexing, or analytical integration rather than executable knowledge models capable of supporting formal reasoning. In enterprise settings, where decision support often depends not only on accessing information but on interpreting constraints, dependencies, and domain semantics across heterogeneous sources, this limitation becomes particularly significant.

To address this challenge, this paper introduces the *Semantic Onboarding* methodology, an ontology-guided approach for transforming heterogeneous organizational artifacts into executable knowledge models. Rather than focusing solely on extracting structured information, the proposed methodology aims at constructing semantically grounded formal representations that can be logically executed and aligned with domain conceptual models. The approach combines LLM-based formalization, schema induction, instance-level knowledge generation, and ontology-guided semantic alignment to progressively transform heterogeneous artifacts into machine-interpretable knowledge bases suitable for reasoning, traceability, and decision support.

The proposed methodology is evaluated using heterogeneous real-world artifacts from a large public financial institution, including 72 service contracts and structured operational database records describing contract life-cycle events such as amendments, renegotiations, penalties, and operational updates. This evaluation investigates the feasibility of constructing ontology-aligned executable knowledge representations from heterogeneous enterprise sources in

a realistic organizational environment.

The main contributions of this work are: (i) a formal ontology-guided methodology for transforming heterogeneous organizational artifacts into executable knowledge models; (ii) a schema induction and semantic alignment process connecting artifact-derived logical representations to domain ontologies; (iii) an executable knowledge engineering approach that enables traceable logical reasoning over enterprise information; and (iv) an empirical evaluation demonstrating the feasibility of the proposed methodology in a real-world enterprise setting involving heterogeneous information sources.

The remainder of this paper is organized as follows. Section 2 presents the foundational concepts related to knowledge representation, ontologies, and LLM-based formalization. Section 3 discusses the related work. Section 4 details the Semantic Onboarding methodology. Section 5 presents the evaluation and experimental results. Finally, Section 6 concludes the paper and outlines directions for future work.

2 Background

This section discusses the role of formal knowledge representation in information systems and the use of ontologies for conceptual modeling. We then recap the distinction between TBox and ABox for knowledge bases and conclude with recent developments in the use of LLMs for knowledge extraction and structuring.

Formal Knowledge representation aims to encode information about a domain in a form that can be interpreted and processed by computational systems. It enables explicit representation of entities, relationships, and constraints, allowing automated reasoning and semantic querying over structured knowledge bases. Logic-based representations have long been used to model domain knowledge in a machine-interpretable form. In such approaches, knowledge is represented as sets of logical statements that can be evaluated through reasoning mechanisms. Description logics and rule-based systems provide formal semantics for representing domain concepts and relationships while supporting inference over implicit knowledge (Baader et al., 2017; Li et al., 2022). These representations form the basis of many current semantic systems, including knowledge graphs and ontology-driven information systems. By structuring domain knowledge as formal models, they enable reasoning tasks such as classification, constraint checking, and semantic querying. Formal knowledge models are widely used in decision-support systems, data integration platforms, and knowledge-driven applications.

Ontologies and Conceptual Modeling provide a formal and explicit specification of the conceptual structure of a domain. They define the vocabulary used to describe domain entities, as well as the relationships and constraints that connect them. In conceptual modeling and knowledge engineering, ontologies serve as semantic frameworks that support knowledge sharing, interoperability, and reasoning (Li et al., 2022). In ontology-based systems, concepts are typically organized in hierarchical structures and connected through semantic relations. These conceptual structures enable the representation of domain knowledge in a way that is both human-understandable and machine-interpretable. Ontologies are therefore widely used in the construction of knowledge graphs, semantic information systems, and intelligent decision-support platforms. Recent work has emphasized the role of ontologies in integrating heterogeneous data sources and guiding knowledge construction pipelines. In knowledge graph construction, for example, ontologies often provide the conceptual schema that governs the extraction and integration of entities and relations from raw data (Huang et al., 2024). By grounding extracted information in a shared conceptual model, ontology-guided systems can ensure semantic consistency across heterogeneous sources.

Knowledge bases in ontology-driven systems are typically divided into two parts: the terminological (TBox) and the assertional components (ABox) (Baader, 2003). The TBox defines the conceptual schema of the domain, including classes, relationships, and logical constraints. In contrast, the ABox contains instance-level assertions describing specific entities and their relationships. The TBox, therefore, captures the structural knowledge of a domain, while the ABox represents concrete facts about individual instances (Bienvenu, 2022). Reasoning systems can use the TBox to infer additional relationships within the ABox, enabling tasks such as instance classification and consistency checking. This separation between schema-level and instance-level knowledge is central to ontology-based knowledge representation. It allows knowledge bases to maintain a stable conceptual structure while continuously incorporating new instance-level information. In many knowledge construction pipelines, the TBox serves as a schema that guides the generation or extraction of ABox facts from heterogeneous data sources.

Finally, Large Language Models (LLM) have recently emerged as a powerful tool for extracting and

structuring knowledge from unstructured and semi-structured data. By leveraging their contextual understanding capabilities, LLMs can identify entities, relations, and events in heterogeneous sources, facilitating the construction of structured knowledge representations (Zhu et al., 2024). Recent research has explored the use of LLMs for knowledge graph construction and semantic information extraction. These approaches consider LLMs to transform natural language artifacts into structured triples or logical representations that can be incorporated into knowledge bases (Liang et al., 2024). This paradigm represents a shift from traditional rule-based extraction pipelines toward generative approaches capable of integrating heterogeneous information sources. Furthermore, ontology-guided LLM pipelines have been proposed to improve the semantic consistency of extracted knowledge. By grounding generation processes in predefined ontological structures, these approaches can constrain model outputs and reduce inconsistencies during knowledge graph construction (Feng et al., 2024). These developments highlight the growing role of LLMs as assistants for knowledge engineering tasks, including ontology construction, schema induction, and instance population.

3 Related Work

Research on transforming heterogeneous information artifacts into structured and executable knowledge representations spans several research areas, including knowledge graph construction, ontology-based knowledge engineering, and logic-based knowledge formalization. LLMs have significantly influenced these research directions by enabling scalable extraction and structuring of information from natural language sources.

Several approaches leverage LLMs to construct knowledge graphs directly from textual artifacts, identifying entities and relationships that can be organized into graph structures (Zhu et al., 2024). These methods have demonstrated promising results in automatically generating triples and relational structures from large document collections. Other studies (Pan et al., 2023) investigate LLM-assisted pipelines for knowledge graph construction supporting entity recognition, relation extraction, and schema induction simultaneously. Such approaches often combine neural extraction methods with symbolic representations in order to improve interpretability and reasoning capabilities.

In parallel, Retrieval-Augmented Generation (RAG) architectures have been widely adopted to improve the relevance of LLM-generated responses (Lewis et al., 2020). These approaches retrieve text segments based on similarity and use them as context for generation, although according to (REF2, 2025), similarity does not necessarily imply relevance to the user's query. Other approaches, such as GraphRAG, incorporate structured representations like knowledge graphs to enhance reasoning (Knollmeyer et al., 2025). While these methods improve multi-hop reasoning and contextual grounding, they remain fundamentally based on similarity-driven retrieval and stochastic generation.

Another emerging line of research explores integrating LLMs with ontology-driven pipelines for knowledge graph population. In these systems, ontological schemes are used to constrain the extraction process, improving semantic consistency and reducing structural inconsistencies in automatically generated graphs (Zhao et al., 2025a). Despite these advances, many existing approaches focus primarily on producing graph-based knowledge representations aimed at storage, integration, or graph traversal. Even ontology-enhanced retrieval approaches improve contextual grounding but do not produce executable knowledge models or support fully deterministic reasoning. In contrast, our work focuses on generating logic-based formal representations that are natively executable and support symbolic reasoning through a logic programming environment.

Ontology-based knowledge engineering has long been used to structure domain knowledge and enable semantic interoperability across heterogeneous information sources. Ontologies provide explicit conceptual models that define entities, relations, and constraints within a domain (Guarino et al., 2009). In recent years, ontologies have played a central role in knowledge graph construction pipelines that guide the extraction and integration of heterogeneous data sources (Keet, 2023). Ontology engineering methodologies have also been applied in regulatory and legal domains to support formal reasoning and compliance verification. Privacy and regulatory ontologies (Palmirani et al., 2018) have been developed to represent legal concepts and support automated reasoning. These approaches demonstrate the potential of ontological models to structure complex normative knowledge. However, ontology-based approaches often assume that instance-level knowledge is already available or manually curated. Our work focuses rather on the earlier stage of knowledge construction, in which formal instance-level representations are generated directly from heterogeneous artifacts and subsequently aligned with ontological models.

Another important line of research concerns the

transformation of natural-language artifacts into executable logical representations. Logic programming languages, such as Prolog, have been widely used to encode formal knowledge models that support automated reasoning. In the legal domain, several systems have explored the use of logic-based representations to formalize statutory rules and regulatory frameworks. For example, the PROLEG system provides a Prolog-based framework for representing legal rules, exceptions, and reasoning structures in a form that can be executed by logic engines (Satoh, 2023). Recent work has also explored integrating LLMs with logic-based knowledge representation frameworks. In particular, a human-in-the-loop workflow has been proposed to transform legal provisions into executable rules by combining LLM-based extraction with expert validation (Zin et al., 2026). At the same time, recent research has begun to introduce structured reasoning capabilities within retrieval-based systems, including graph-enhanced pipelines and hybrid reasoning architectures (Zhao et al., 2025b).

While all these approaches improve reasoning performance, they remain dependent on retrieval and do not provide fully deterministic or executable logic representations. This research work builds upon these research directions to address these gaps.

4 Semantic Onboarding Methodology

We propose a structured process for constructing ontology-enriched formal knowledge models from heterogeneous information sources. Let $S = \{s_1, s_2, \ldots, s_n\}$ be a collection of information artifacts originating from heterogeneous sources, including documents, database tables, spreadsheets, emails, and other structured or semi-structured data representations. Let $\Omega = \{O_1, O_2, \ldots, O_k\}$ be a set of ontologies selected to support the semantic enrichment of the generated knowledge models. The objective of Semantic Onboarding is to transform S into a set of formal models M suitable for automated reasoning. Formally, the process may be defined as follows:

$$SO : (S, \Omega) \rightarrow M$$

where SO denotes the Semantic Onboarding transformation, S is the source collection, Ω is the ontology set, and M is the resulting collection of ontology-enriched formal knowledge models. The proposed methodology is structured into seven phases:

Initial Formal Model Generation The first stage of the method consists of selecting representative samples from the heterogeneous information sources that compose the artifact collection. Let $S_{sample} \subseteq S$ denote a set of sampled artifacts obtained from each source under consideration. When multiple sources are involved, such as documents and database tables, the sample set contains representative artifacts from each source type. These sampled artifacts are used to induce candidate formal representations of the source structures. For each sampled artifact $s \in S_{sample}$, a LLM generates one or more candidate logic-based representations. Let $M_c = \{m_1, m_2, ..., m_r\}$ be the set of candidate formal models obtained from the sampled artifacts.

The generated models are expressed in Prolog. This choice is motivated by both representational and operational considerations. From a representational perspective, Prolog provides a declarative and transparent formalism in which entities, relations, and constraints can be explicitly represented as symbolic structures. From an operational perspective, Prolog supports executable reasoning through rule-based inference and query answering. In contrast to purely descriptive representations, Prolog models can be used directly as machine-interpretable knowledge bases. Moreover, the fact-and-rule structure of Prolog aligns naturally with ontology-informed conceptual modeling, since semantically structured concepts and relations can be encoded in a way that remains both computationally executable and logically inspectable.

Human-Guided Representative Model Selection Candidate models are evaluated by a human expert who determines which of them best captures the semantic structure of the source artifacts. This model m serves as the reference formalization for the source type under analysis. The purpose of this stage is not merely quality control, but semantic anchoring. By selecting the model that best reflects the intended conceptualization of the source, the expert establishes the basis for the subsequent schema-induction step.

TBox Construction Once the representative model has been selected, the method derives a structural schema from its logical vocabulary. This schema plays a TBox-like role in the overall process, capturing the predicates, arities, and associated metadata that define the formal representation's structural backbone. Let $P = \{p_1, p_2, \ldots, p_q\}$ be the set of predicates found in m. The TBox is constructed by extracting these predicates and their corresponding metadata deterministically, without using a language model. The resulting TBox establishes the expected representational structure for all subsequent instance-level encodings derived from the same source type. Formally, let T denote the induced TBox $T = schema(m^*)$ where $schema(\cdot)$ is a deterministic extraction function over the selected representative model.

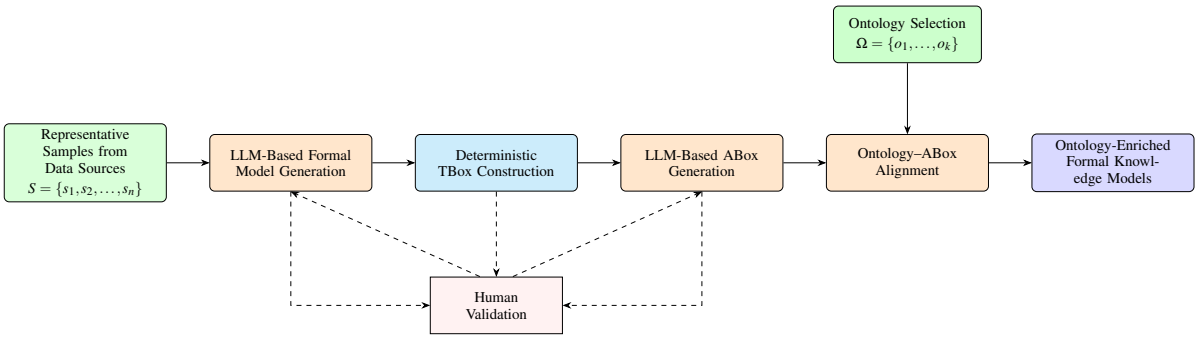


Figure 1: Semantic Onboarding methodology for constructing ontology-enriched executable knowledge bases.

ABox Generation Given the induced TBox T, the method proceeds to generate instance-level formal representations for all artifacts in the source collection. For each source artifact $s_i \in S$, a large language model produces an ABox A_i constrained by the predicate vocabulary and structural expectations defined in T. Thus, the output of this phase is a set of ABoxes, where each A_i corresponds to the formal encoding of artifact s_i: $A = \{A_1, A_2, \ldots, A_n\}$. The use of a shared TBox ensures structural consistency across all generated ABoxes. Rather than allowing each instance to induce its own ad hoc representation, the method constrains generation through a common symbolic schema, thereby reducing representational drift and increasing comparability among the resulting models.

Iterative Human Validation As we can see in Figure 1, all the generated artifacts, namely Initial Formal Models, TBox, ABoxes, and Ontology-Enriched ABoxes, are inspected by a human expert who evaluates their structural adequacy, semantic fidelity, and alignment with the intended conceptualization of the source. This validation stage supports iterative refinement of the onboarding process. If significant inconsistencies or recurring representational deficiencies are identified, the process may return to the TBox construction phase or to the ABox generation phase. In practice, this means that schema induction and instance generation are not strictly linear, but may be revisited until a satisfactory level of semantic adequacy is achieved.

Ontology Selection After the construction of the base formal models, a human expert identifies the ontology set Ω to be used in the enrichment phase. Importantly, the method assumes that multiple ontologies may be necessary to adequately represent complementary perspectives over the same source artifacts. These ontologies may belong to different layers of conceptualization. For example, a foundational ontology may provide high-level categories for events, roles, and relations, while domain ontologies may capture service contracts, and analytical ontologies may introduce value, risk, or security-related concepts. Therefore, the enrichment phase is explicitly multi-ontological.

Ontology-Guided Semantic Enrichment The final phase of the methodology aligns each generated ABox with the selected ontologies. For each pair (A_i, O_j) such that $A_i \in A$ and $O_j \in \Omega$, an alignment process is executed with the support of an LLM. This process identifies semantic correspondences between the predicates and individuals in the ABox and the classes and relations defined in the ontology. Formally, the enrichment function can be defined as stated below, where $A_i^{(j)}$ is the enriched version of A_i after alignment with ontology O_j.

$$enrich : (A_i, O_j) \rightarrow A_i^{(j)}$$

When multiple ontologies are used, the enrichments are accumulated over the same instance-level model. Thus, for each source artifact, the final enriched ABox can be expressed as:

$$A_i' = enrich_{O_k}(\ldots enrich_{O_2}(enrich_{O_1}(A_i))\ldots)$$

The final collection of enriched models is $M = \{A_1', A_2', \ldots, A_n'\}$. This process preserves the cardinality of the original source collection: if the input contains n artifacts, the final output also contains n instance-level formal models, now semantically enriched through ontology alignment.

5 Evaluation

We evaluate the proposed Semantic Onboarding methodology using a real-world enterprise case study involving heterogeneous contractual artifacts from a large public financial institution. The evaluation investigates the construction, semantic enrichment, and executable reasoning capabilities of the resulting formal knowledge models, using prior architectures as comparative baselines.

We preserve confidentiality by anonymizing organization names and contract identifiers. To ensure reproducibility, all source code, input datasets, and evaluation artifacts are available in the anonymous replication repository at https://anonymous.4open.science/.

Evaluation Objectives The evaluation investigates whether the proposed Semantic Onboarding methodology can systematically transform heterogeneous enterprise artifacts into executable ontology-aligned knowledge models suitable for reasoning and decision support. More specifically, the evaluation addresses the following research questions:

- **RQ1:** Can heterogeneous enterprise artifacts be consistently transformed into structurally coherent formal knowledge representations?
- **RQ2:** Does ontology-guided semantic enrichment increase the semantic expressiveness of the generated knowledge models beyond purely structural extraction?
- **RQ3:** Do the resulting executable knowledge models support deterministic logical reasoning over enterprise information?
- **RQ4:** How does the proposed methodology differ architecturally from retrieval-based decision-support approaches operating in the same domain?

Dataset and Setting The evaluation was conducted using heterogeneous real-world contractual artifacts from a large public financial institution. The contractual dataset comprises 72 real-world IT service contracts together with structured operational records describing amendments, renegotiations, penalties, and related lifecycle events. Semantic enrichment was performed using the service contract ontology proposed by (Griffo et al., 2021), grounded in UFO. The resulting executable knowledge models were represented and queried in SWI-Prolog.

Artifact Validation To assess the quality of the generated knowledge artifacts, the evaluation employed a multi-role expert validation process involving contract managers, cybersecurity risk specialists, and data engineers. Structural validation focused on the consistency of the generated formal representations. All 72 generated ABoxes, together with the induced schema (TBox), were inspected by data engineers using a formal checklist designed to verify predicate consistency, metadata completeness, structural coherence, and conformity with the expected schema.

Semantic validation focused on ontology adequacy and domain interpretability. Contract managers and cybersecurity risk specialists inspected the ontology-enriched representations to verify whether the generated semantic relations accurately reflected known contractual structures, obligations, roles, and cybersecurity-relevant relationships typically found in IT service contracts. This validation protocol ensured that the generated knowledge models were assessed not only for structural consistency, but also for semantic adequacy with respect to the intended enterprise domain.

Executable Reasoning Validation To evaluate the reasoning capabilities of the generated executable knowledge models, we reused the benchmark competency questions defined in the baseline architecture presented in (REF1, 2024), enabling direct comparison between retrieval-based and ontology-grounded reasoning paradigms.

The benchmark comprises 14 competency questions covering both document-level contractual information and operational contract-management data. These questions were operationalized as executable Prolog queries with LLM assistance and executed over the ontology-aligned knowledge models produced by the Semantic Onboarding pipeline. The resulting answers were independently inspected by contract managers and domain specialists to verify semantic adequacy, contractual fidelity, and practical usefulness for enterprise decision-support tasks.

Representative Formal Model As evidence of the artifact construction process, we present representative formal models produced during the evaluation. The first stage of the Semantic Onboarding process consists of generating candidate formal models from sampled artifacts of the source datasets.

Listing 1: Example of clause-level facts generated during onboarding

```
contract(contract_001_2026).
contract_metadata(contract_001_2026, contracting_party,
      a_bank).
contract_metadata(contract_001_2026, contractor, b_tech).
contract_clause_fact(contract_001_2026, clause_duration,
      contract_term, '36 months', clause_text).
contract_clause_fact(contract_001_2026, clause_duration,
      maximum_extension_period, '60 months', clause_text).
contract_clause_fact(contract_001_2026, clause_duration,
      extension_response_deadline, '5 business days',
      paragraph_1).
contract_clause_fact(contract_001_2026, clause_duration,
extension_notice_deadline, '90 days', paragraph_2).
...
```

Since the methodology relies on an LLM to generate these representations, the resulting models are not deterministic. Even when processing artifacts with similar structure, organization, and terminology, the language model may produce different logical representations of the same underlying information. Consequently, multiple candidate models emerge from the

sampled artifacts, reflecting alternative formalizations of the source content.

Induced Schema (TBox) Based on the representative model selected by a human expert, the methodology generates a schema that describes the structural vocabulary of the extracted knowledge base. This schema is represented as a TBox containing predicate signatures and class definitions. The generated schema serves as a structural template for generating instance-level models. A fragment of the induced schema is shown below.

Listing 2: Excerpt of the induced schema (TBox)

```
class(contract).
predicate_signature(contract,1).
predicate_signature(contract_metadata,3).
predicate_signature(contract_clause,3).
predicate_signature(contract_clause_fact,5).
...
```

The TBox ensures that all instance-level models share a consistent structural vocabulary. As a result, multiple contracts processed through the pipeline can be represented under a common schema.

Instance-Level Knowledge (ABox) Using the generated schema, the pipeline produces instance-level knowledge models for each contract artifact. These models are represented as ABoxes containing logical assertions describing specific contractual structures. For Contract 002/2026, the generated ABox includes facts describing the contractual roles and clause structures.

Listing 3: Example of ABox generated during onboarding

```
contract(contract_002_2026).
contract_metadata(contract_002_2026, contracting_party,
      a_bank).
contract_metadata(contract_002_2026, contractor, d_tech).
...
contract_clause_fact(contract_002_2026,
      clause_contract_guarantee,
document_submission_deadline, '10 business days',
      clause_text).
contract_clause_fact(contract_002_2026,
      clause_contract_guarantee,
guarantee_amount, '133000.00', clause_text).
contract_clause_fact(contract_002_2026,
      clause_contract_guarantee,
guarantee_type, 'cash deposit', item_1).
contract_clause_fact(contract_002_2026,
      clause_contract_guarantee,
guarantee_type, 'performance bond insurance', item_2).
...
```

The generation of ABox assertions is guided by the previously induced TBox, ensuring that the produced facts conform to the predicate signatures and metadata structures defined in the schema. In particular, the attributes associated with each clause correspond to the set of metadata elements anticipated by the TBox. At the same time, the methodology preserves the completeness of the extracted information. As mentioned earlier, if a contractual artifact contains metadata that is not explicitly defined in the schema, the corresponding fact is still recorded in the ABox and explicitly marked as not belonging to the original schema, using `contract_metadata_raw`, as shown below.

Listing 4: Example of ABox generated during onboarding

```
contract_metadata_raw(contract_002_2026, '
     internal_reference_code', '016/2025', reference).
```

This mechanism prevents information loss while maintaining a clear distinction between schema-conforming and schema-extending knowledge, and ultimately opens the door to schema evolution and open-world modeling.

Ontology-Guided Alignment and Semantic Enrichment In the final stage of Semantic Onboarding, the generated ABoxes are semantically enriched through alignment with one or more selected ontologies. The objective of this phase is to transform structurally consistent formal representations into semantically grounded knowledge models by establishing explicit correspondences between extracted predicates and entities and the conceptual structures defined in the selected ontologies.

Listing 5: Excerpt of Ontology-Aligned ABox

```
plays_role(a_bank, service_customer_role, contract_002_2026
      ).
plays_role(d_tech, hired_service_provider_role,
      contract_002_2026).

clause_of(clause_service_levels, contract_002_2026).

contract_clause_fact(clause_service_levels,
      service_execution,
    b_tech, execute_services_according_to_standards).

legal_relation_instance(clause_service_levels, duty_to_act,
    b_tech, execute_services_according_to_standards).

legal_relation_instance(clause_service_levels,
      right_to_action,
    a_bank, require_service_execution_according_to_standards
         ).
```

The alignment process operates over both the generated ABox vocabulary and the ontology schema. For each predicate, entity, or relation represented in the ABox, candidate semantic correspondences are

Table 1: Comparison with baseline (REF1, 2024)

Aspect	Baseline (RAG + Agents)	Proposed Methodology
Answer generation	Stochastic (LLM-based generation)	Deterministic (logical query evaluation)
Reproducibility	Not guaranteed	Fully reproducible
Retrieval mechanism	Similarity-based (vector search)	Not required (explicit knowledge models)
Relevance vs similarity	Similarity used as proxy for relevance	Explicit semantic interpretation
Knowledge representation	Implicit (text + embeddings)	Explicit (formal logical models)
Reasoning capability	Limited (generation-based)	Logical inference over ABox/TBox
Explainability	Indirect (text-based justification)	Direct (traceable facts and rules)
Semantic structure	Implicit in text	Explicit (ontology-guided alignment)
Handling ambiguity	Dependent on prompt and context	Reduced via formal semantics

identified with ontological concepts, roles, and relations. This mapping is operationalized with LLM assistance, which considers the predicate semantics, the contextual meaning of the source artifact, and the conceptual definitions available in the ontology. When ambiguities arise, expert validation ensures semantic adequacy and conceptual consistency.

Importantly, ontology-guided enrichment may introduce semantic interpretations that go beyond direct structural correspondence. Structural predicates extracted from contractual artifacts may be reinterpreted as explicit legal roles, obligations, rights, or normative relations defined in the ontology, thereby adding semantic commitments not directly present in the original structural representation.

In our evaluation, the selected ontology was the service contract ontology proposed by (Griffo et al., 2021), which models contractual concepts, roles, commitments, and legal relations grounded in the Unified Foundational Ontology (UFO). In the example shown in listing 5, clause-level operational facts describing service execution are aligned with ontological concepts such as `duty_to_act` and `right_to_action`, while contractual actors are interpreted through role assignments using `plays_role`. Thus, `b_tech`, as the hired service provider, bears the duty to execute services according to contractual standards, whereas `a_bank`, as the service customer, holds the corresponding right to require such execution. As a result, the knowledge base evolves from a purely structural representation of extracted facts into an ontology-grounded executable model that explicitly captures reciprocal legal and semantic relations between contractual parties.

Reasoning over ontology-aligned executable knowledge models Since the resulting artifacts are represented as Prolog facts, they can be directly executed, queried, and reasoned over using standard logical inference mechanisms. The ontology-aligned ABoxes produced by the Semantic Onboarding pipeline therefore constitute an executable knowledge base that can be explored through Prolog queries. This executable reasoning capability was also used to operationalize the 14 benchmark competency questions adopted in the baseline evaluation.

In our evaluation, the resulting knowledge base supports analytical queries over contractual roles and legal relations. For example, it becomes possible to retrieve all contractual duties associated with a specific party. The following query identifies all duties assigned to the service provider B_Tech.

Listing 6: Query and resulting duties derived from the ontology-aligned ABox

```
?- legal_relation_instance(_, duty_to_act, b_tech, Action).

Action = provide_etl_services (Clause 1 - Object).
Action = execute_services_according_to_standards (Clause 4
     - Service Levels).
Action = present_fiscal_document (Clause 7 - Payment).
Action = provide_contractual_guarantee (Clause 10 -
     Contractual Guarantee).
Action = maintain_confidentiality_of_information (Clause 13
      - Confidentiality).
Action = maintain_data_confidentiality (Clause 14 - Data
     Protection).
...
```

The query returns the set of contractual duties associated with the service provider. In the analyzed contract, these duties correspond to actions derived from several clauses, including service execution, payment procedures, contractual guarantees, ethical conduct, and data protection obligations. More complex reasoning tasks can also be performed by combining ontology-level concepts with instance-level knowledge. For instance, it is possible to identify all contractual relations associated with a specific clause.

Comparison with Baseline We compare the proposed methodology with the retrieval-based architecture presented in (REF1, 2024), which combines Retrieval-Augmented Generation (RAG), text-to-SQL, and agent-based orchestration to support question answering in the same contractual domain. Unlike the proposed approach, the baseline depends on similarity-based retrieval mechanisms and genera-

tive answer construction, which may introduce variability across executions and limit reproducibility, traceability, and formal consistency.

Listing 7: Querying legal relations from ontology-aligned service contracts

```
?- legal_relation_instance(clause_service_levels,
     RelationType, Agent, Action).

RelationType = duty_to_act, Agent = b_tech,
Action = execute_services_according_to_standards.

RelationType = right_to_action, Agent = a_bank,
Action = require_service_execution_according_to_standards.

RelationType = power, Agent = a_bank,
Action = apply_price_reduction_index.

RelationType = subjection, Agent = b_tech,
Action = be_subject_to_price_reduction.

RelationType = power, Agent = a_bank,
Action = apply_penalties_for_non_compliance.

RelationType = subjection, Agent = b_tech,
Action = be_subject_to_penalties.
```

By contrast, Semantic Onboarding produces explicit ontology-aligned executable knowledge models that support deterministic logical inference over contractual facts, roles, obligations, and legal relations. Table 1 summarizes the principal architectural differences between both approaches.

Using the same benchmark adopted by the baseline, the generated executable knowledge models successfully supported all 14 competency questions, as independently validated by contract managers and domain specialists. Beyond reproducing the benchmark analytical capabilities, the ontology-guided enrichment introduced additional reasoning capabilities unavailable in retrieval-based architectures, including direct querying over explicit legal relations, normative obligations, role semantics, and formally traceable contractual dependencies.

Evaluation Findings Overall, the evaluation provides affirmative evidence for the research questions introduced at the beginning of this section. First, the successful generation and expert validation of the induced structural schema and all 72 executable ABoxes indicate that heterogeneous enterprise artifacts can be systematically transformed into structurally coherent formal knowledge representations. Second, the ontology-guided enrichment process substantially increases the semantic expressiveness of the resulting models by introducing explicit concepts such as contractual roles, legal obligations, rights, and normative relations that are not directly available in the original source artifacts or in purely structural formalizations. Third, the ontology-aligned executable knowledge models demonstrated deterministic logical reasoning capabilities, both through direct Prolog-based analytical queries and through successful support of all 14 benchmark competency questions adopted from the baseline architecture. Finally, the comparison with retrieval-based architectures highlights a fundamentally different reasoning paradigm, in which explicit formal semantics replace similarity-based retrieval and stochastic answer generation, enabling reproducibility, traceability, and richer ontology-grounded analytical capabilities.

6 Conclusions and Future Work

This paper introduced Semantic Onboarding, an ontology-guided methodology for systematically transforming heterogeneous enterprise information artifacts into executable formal knowledge models. Unlike retrieval-centric architectures that rely on similarity-based access and stochastic answer generation, the proposed approach constructs explicit ontology-aligned representations that support deterministic logical reasoning, semantic traceability, and reproducible analytical querying.

The evaluation conducted over 72 real-world IT service contracts and structured operational records from a large public financial institution demonstrated the feasibility of the methodology in a realistic enterprise setting. Expert validation confirmed the structural consistency and semantic adequacy of the generated knowledge artifacts, while the ontology-aligned executable models successfully supported all 14 benchmark competency questions adopted from the baseline architecture. These results indicate that Semantic Onboarding provides a viable knowledge engineering approach for semantically grounded enterprise reasoning and decision-support scenarios.

Future work will investigate extensions in three directions. First, we intend to support controlled schema evolution, allowing the induced structural schema to be incrementally refined as new artifacts introduce recurring attributes, relations, or structural patterns. Second, we plan to explore broader open-world reasoning capabilities for situations involving incomplete or partially observable enterprise knowledge. Third, we intend to evaluate the methodology across additional enterprise domains and alternative ontological models, assessing its generalizability beyond service contracts and reducing dependence on domain-specific expert supervision during semantic alignment.

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